From Conflict to Human Capital?
Evidence from Colombia's Peace Process

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Abstract

Can the end of armed conflict alter the transition from secondary school to higher education? I link 6.22 million regular, second semester Saber 11 examinees from 2007 to 2018 to Saber Pro records through 2024 and combine them with a municipality-year panel of conflict events. Historical FARC municipalities experience a large relative decline in conflict after negotiations begin: the pooled conflict-index effect is -0.42 standard-deviation units relative to historical ELN municipalities, with stable pre-trends. Yet the same FARC versus ELN design yields near-zero effects on Saber 11 participation and achievement, progression to Saber Pro, and score value added. PDET territories display a different pattern. Progression to Saber Pro within six years initially falls when negotiations begin and reverses after the 2014 unilateral ceasefire, reaching a 3.15 percentage-point relative increase by 2018. Saber 11 participation also rises, while relative scores fall and progression becomes slightly slower. Timing estimates at four-, five-, and six-year horizons reproduce the initial dip, but credit use and a semester-based delay or interruption proxy do not provide direct support for an uncertainty mechanism. The evidence therefore rejects a broad, automatic human-capital peace dividend. It instead shows that a sizable security dividend can coexist with weak educational responses, while access expands only in the territories later prioritized for complementary policy.

Keywords: peace agreements, armed conflict, education, higher education, Colombia, human capital
JEL codes: D74, I21, I23, O15

1 Introduction

The end of a civil conflict can transform educational investment. Lower violence may reduce the risk of traveling to school, weaken incentives for recruitment into armed groups, and lengthen the horizon over which families value schooling. Peace can also open local labor markets, attract institutions, and make higher education newly feasible. None of these effects, however, follows mechanically from disarmament. A territory emerging from conflict may still lack schools, universities, roads, credit, and effective public institutions. The same reduction in violence can expand participation among students who were previously excluded while lowering average test performance through composition. A peace process may therefore increase educational opportunity without immediately increasing measured learning.

Colombia offers an unusually informative setting in which to study this distinction. Formal negotiations between the national government and the Revolutionary Armed Forces of Colombia, the FARC, began in 2012. The FARC declared an indefinite unilateral ceasefire in December 2014, and the parties signed a final agreement in November 2016. These events separated political negotiation from a credible reduction in hostilities and from the later implementation of territorial policy. They also generated sharp spatial variation. Some municipalities had long histories of FARC presence, while others were exposed to the National Liberation Army, the ELN, which did not participate in the FARC agreement. In 2017, the government additionally designated 170 conflict affected, poor, rural municipalities for the Programas de Desarrollo con Enfoque Territorial, or PDET (Colombia, 2017; United Nations, 2017).

This paper asks whether the process of peace changed the path from the end of secondary school to the final stage of university. I build a linked administrative data set that begins with students taking Saber 11, Colombia's secondary school exit examination, during the regular second semester session. I then search for those students in Saber Pro, the national examination taken by university students who have completed at least 75 percent of their degree credits. Because taking Saber Pro is a degree requirement, appearance in the examination is a strong indicator that a student has progressed close to university completion, but it is not a formal graduation record (ICFES, 2026). I use a six year window so that every secondary school cohort in the analysis has comparable follow-up through the 2024 Saber Pro files.

The empirical analysis combines two designs that answer related but distinct questions. The first compares students from PDET municipalities with students from non-PDET municipalities. This design has statistical power and captures the places at the center of the post-conflict policy agenda, but treatment status was assigned after the peace agreement using poverty, rurality, institutional weakness, illicit crops, and conflict exposure. PDET and non-PDET municipalities are consequently quite different in levels. The second design compares municipalities with historical FARC presence with those with historical ELN presence. This comparison follows the logic of Fajardo-Steinhäuser (2024): both groups experienced insurgent conflict, but only FARC territories were directly exposed to the 2014 ceasefire and the subsequent agreement. The FARC and ELN samples are much more similar in pre-period educational achievement, although the ELN control group is small.

Four findings organize the paper. First, the peace process generated a real security dividend in its intended territories. Historical FARC municipalities experience a pooled decline of 0.42 standard-deviation units in a core conflict index relative to historical ELN municipalities after 2012. The corresponding log event-rate estimate is -0.66. Pre-treatment coefficients are jointly insignificant, and the event path turns persistently negative after negotiations begin. This first stage matters because it establishes that the educational null is not produced by a peace process that failed to reduce the forms of violence measured in the data.

Second, the cleaner FARC versus ELN educational comparisons remain close to zero despite that conflict decline. The pooled post-2012 difference in the six-year Saber Pro outcome is 0.11 percentage points with a standard error of 0.81. FARC and ELN municipalities also show no differential improvement in Saber 11 mathematics, verbal performance, or residual value added between Saber 11 and Saber Pro. The contrast between a large conflict reduction and small education estimates is the paper's central result: security improved, but human-capital accumulation did not automatically follow.

Third, PDET territories show a striking but less clean access pattern. Relative to 2011, the probability of appearing in Saber Pro within six years falls by 1.17 percentage points in 2012, turns positive in 2014, and rises to 3.15 percentage points by 2018. Saber 11 participation also increases, while relative mathematics and verbal scores decline and conditional time to Saber Pro becomes modestly longer. This quantity and quality divergence is consistent with an extensive-margin response in territories receiving political attention and complementary programs, rather than a general increase in educational productivity.

Fourth, the new uncertainty tests narrow the interpretation of the initial PDET dip. Event studies for reaching Saber Pro within four, five, and six years all decline around the start of negotiations and recover after the ceasefire. However, the data do not observe applications or first enrollment. Among students who reach university-level Saber Pro, there is no robust PDET effect on credit financing, excess elapsed time relative to reported semester, or a delay and interruption proxy. The timing pattern is compatible with families waiting for credible peace, but the direct proxies do not establish uncertainty as the mechanism.

The paper contributes to research on conflict, post-conflict recovery, and higher education. Evidence from Tajikistan, Guatemala, Peru, and Colombia shows that armed violence can reduce enrollment, attainment, and achievement (Shemyakina, 2011; Chamarbagwala and Morán, 2011; León, 2012; Rodríguez and Sánchez, 2012; Gómez Soler, 2016). The present analysis shifts attention from wartime destruction to the conditions under which declining violence produces recovery. It complements evidence that Colombia's ceasefire improved some school outcomes (Namen et al., 2021) and that coverage can expand while relative achievement gaps widen (Montaño-Bardales and Palacios, 2025; Avendaño Santiago, 2021). It also connects post-conflict education to work on uncertainty, sequential schooling choices, and completion risk (Kazianga, 2012; Stange, 2012; Athreya and Eberly, 2021). Finally, it speaks to Colombia's higher-education literature, which emphasizes supply responses, information, financial aid, and credit constraints (Carranza and Ferreyra, 2019; Londoño-Vélez et al., 2020; Melguizo et al., 2016).

The central lesson is deliberately narrower than the claim that peace had no educational impact. Peace appears to have relaxed an access constraint in the places Colombia later prioritized, but it did not reliably improve achievement, speed, or value added. The results suggest that security is an input into human capital formation, not a substitute for educational capacity.

2 Related Literature and Conceptual Framework

The first relevant literature studies how conflict destroys human capital. Violence can close schools, displace families and teachers, increase child labor, expose students to trauma, and reduce the expected return to geographically specific investments. Cross-country evidence associates civil war with lower educational expenditure and enrollment, while microeconomic studies document persistent schooling losses for cohorts exposed to conflict in Tajikistan, Guatemala, and Peru (Shemyakina, 2011; Chamarbagwala and Morán, 2011; León, 2012). In Colombia, conflict exposure is associated with lower enrollment, more child labor, and weaker Saber 11 performance (Rodríguez and Sánchez, 2012; Gómez Soler, 2016, 2017). Justino (2016) emphasizes that these effects combine demand restrictions faced by families with supply restrictions in schools and the state.

The post-conflict question is conceptually different. Removing violence relaxes one constraint, but damaged infrastructure, thin markets, low state capacity, and weak educational supply may remain. A peace dividend should therefore be decomposed into a security first stage and downstream behavioral responses. Colombian evidence already suggests such complementarity. The FARC ceasefire reduced violence and was followed by changes in school outcomes and entrepreneurship, but broad economic activity remained weak where the state did not enter effectively (Namen et al., 2021; Bernal et al., 2024; Fajardo-Steinhäuser, 2024). The new conflict panel allows this paper to show the first stage directly and then ask whether the same spatial contrast changed education.

Uncertainty provides one possible bridge between violence and schooling decisions. Education is costly, partially irreversible, and pays off with delay. When households face uninsured income or security risk, they may postpone enrollment or favor liquid resources over long-horizon human-capital investments (Kazianga, 2012). College decisions are also sequential: enrollment has option value because students learn about aptitude and costs before deciding whether to continue (Stange, 2012). Completion risk can consequently dampen enrollment even when the college premium is high (Athreya and Eberly, 2021). In Colombia, credit and aid can sharply increase enrollment, especially for low-income students, while institutional supply responds to demand (Melguizo et al., 2016; Londoño-Vélez et al., 2020; Carranza and Ferreyra, 2019). These mechanisms motivate the timing, financing, and interruption proxies in Code 12.

The predictions are deliberately separated. If negotiation uncertainty temporarily delayed investment, early post-2012 cohorts should be less likely to reach Saber Pro at short horizons, with recovery after the ceasefire. If liquidity constraints were central, credit or scholarship financing could change among later university examinees. If interrupted or delayed trajectories were important, elapsed time should rise relative to the semester reported at Saber Pro. None of the latter variables is a direct enrollment history, so the paper treats them as mechanism diagnostics rather than additional causal outcomes.

3 The Colombian Peace Process and Education

3.1 From negotiation to ceasefire and implementation

The FARC emerged in the 1960s and became Colombia's largest guerrilla organization. Its territorial presence was concentrated in rural areas where state authority, road access, legal markets, and public services were often weak. Armed conflict affected education through direct violence, displacement, recruitment, teacher turnover, school closure, and the opportunity cost created by illicit and armed activity. Earlier Colombian evidence associates conflict exposure with lower enrollment, more child labor, and weaker school achievement (Rodríguez and Sánchez, 2012; Gómez Soler, 2016, 2017).

The peace process unfolded in stages. Public negotiations began in 2012, but negotiation itself did not end fighting. The more consequential break in security occurred on December 20, 2014, when the FARC initiated an indefinite unilateral ceasefire. Violence attributable to the FARC subsequently fell sharply. A bilateral ceasefire followed in 2016, and the revised Final Agreement was signed on November 24 and ratified by Congress later that month. The agreement entered into force on December 1, 2016 (United Nations, 2016, 2017). This sequence matters for educational choices. A family may react little to talks that can fail, but substantially to an observed and persistent reduction in violence.

The agreement was ambitious. It combined disarmament and political incorporation with rural reform, crop substitution, victim reparations, and territorial development. Yet implementation started from a low-capacity equilibrium. Former FARC municipalities often had limited fiscal resources and thin public service networks. Other armed groups could enter territories vacated by the FARC, and political uncertainty rose after the first version of the agreement was rejected in the October 2016 referendum. The end of one armed relationship therefore did not instantly produce a complete or uniform peace.

3.2 PDET territories and the educational margin

Decree Law 893 of 2017 created the PDET as the territorial vehicle for implementing rural reform in the areas most affected by conflict. It selected 170 municipalities organized into 16 subregions. Selection reflected armed conflict, poverty, institutional weakness, rural conditions, and illicit economies (Colombia, 2017). These municipalities are substantively important because they concentrate the population for whom a peace dividend was most needed. They are not, however, a treatment group created by random or quasi-random assignment.

The distinction between security and policy is central to interpretation. A PDET event study around 2012 or 2014 captures the evolving relative trajectory of territories later selected for post-conflict investment. It may reflect the ceasefire, expectations, local policy, and differential recovery. It should not be described as the causal effect of PDET designation before 2017. By contrast, the FARC versus ELN design more directly isolates exposure to the FARC ceasefire, conditional on the assumption that historical ELN municipalities provide a valid counterfactual.

Higher education is a useful outcome in this setting because it requires a sequence of investments. Students must complete secondary school, take Saber 11, enroll in a professional program, persist through most required credits, and take Saber Pro. A change in near-completion within six years therefore captures more than a short-lived enrollment response. At the same time, the measure can move through supply, demand, migration, financing, or selection. The analysis treats it as an integrated educational progression outcome and uses secondary school participation and scores to clarify the margin of adjustment.

4 Data and Measurement

4.1 Linked examination records

The source data are administrative examination records from the Colombian Institute for the Evaluation of Education, ICFES. Saber 11 is the national examination taken near the end of secondary school. The analysis retains regular students who present the test in the second semester of each year. This restriction is substantive: the paper is designed to follow recent high-school graduates, and the second-semester session is the standard examination period for students finishing grade 11 in Colombia's traditional academic calendar. Restricting attention to this session makes the starting cohort more homogeneous and makes elapsed time to higher education interpretable as a post-secondary trajectory rather than a mixture of recent graduates, adult examinees, and off-cycle test takers.

The resulting analysis sample contains 6,221,403 recent school leavers from the 2007 to 2018 Saber 11 cohorts. Student identifiers are used to link these records with Saber Pro through 2024. A total of 1,999,396 students, or approximately 32 percent, appear at some point in Saber Pro. The main outcome equals one when the first observed Saber Pro record occurs within six years of Saber 11. This condition is met by 1,274,032 students, approximately 20 percent of the analysis sample.

Saber Pro is designed for students near the end of professional university programs. ICFES defines eligible examinees as those who have completed at least 75 percent of their program credits, and the examination is a graduation requirement (ICFES, 2026). I therefore refer to the main outcome as "Saber Pro within six years" or "near-completion within six years." Calling it university graduation would overstate what the administrative record directly observes. Technical and technological programs tested through Saber TyT are not incorporated because comparable pre-peace records are unavailable.

All cohorts have a full six year opportunity to appear in Saber Pro by 2024. The use of a fixed horizon avoids comparing older cohorts with long follow-up to recent cohorts with mechanically shorter exposure. Figure 1 shows that unconditional near-completion declines in recent cohorts in all groups. This broad decline makes cohort fixed effects essential and cautions against reading raw time series as policy effects.

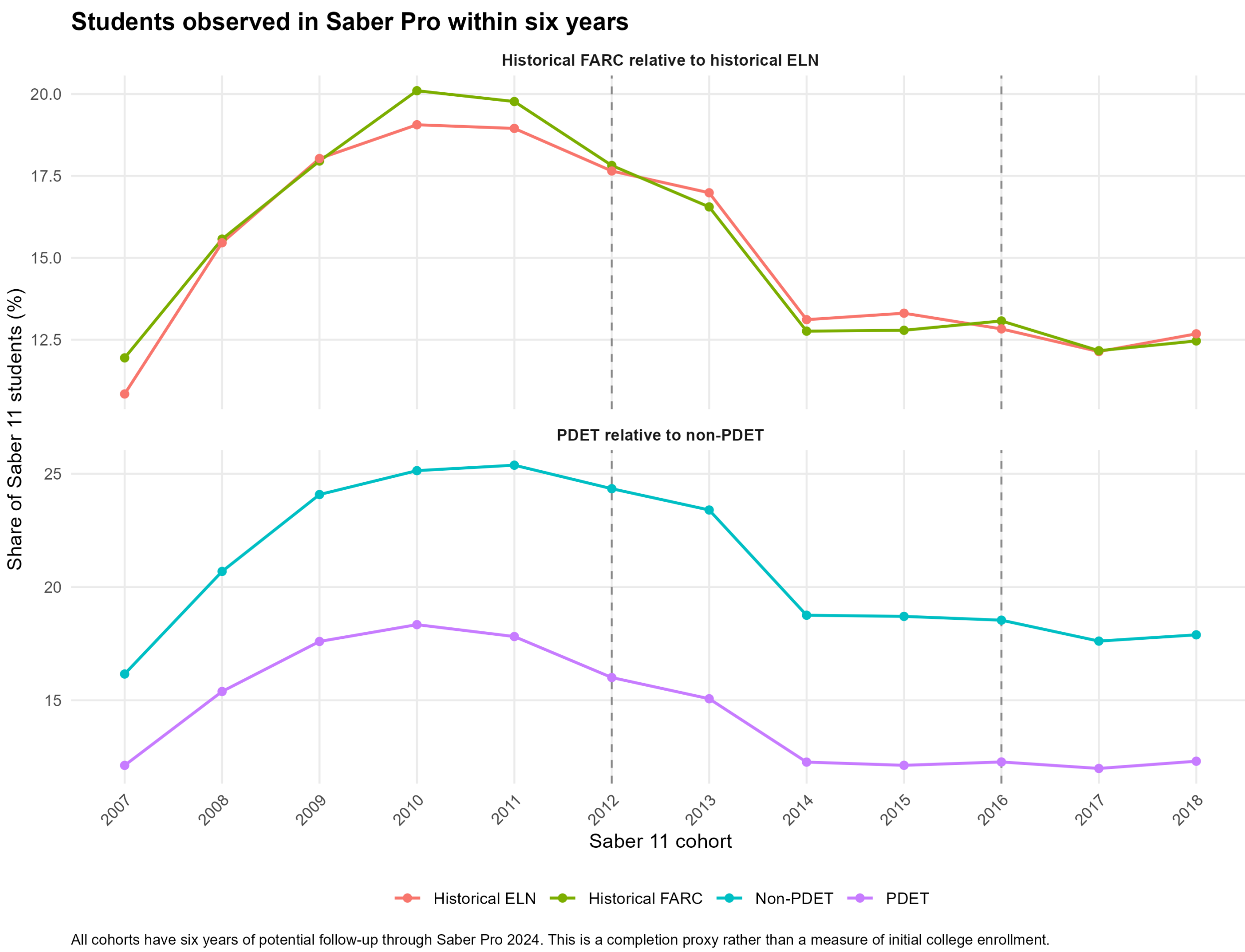


Figure 1: Saber Pro within six years by Saber 11 cohort
Notes: Each point is the share of regular, second semester Saber 11 examinees observed in Saber Pro within six years. Left panel compares PDET and non-PDET municipalities. Right panel compares municipalities with historical FARC and ELN presence.

4.2 Treatment definitions and baseline comparability

The first treatment indicator equals one if the student's municipality of residence at Saber 11 is among the 170 PDET municipalities. The comparison group contains the remaining observed municipalities. The second treatment indicator equals one for municipalities with historical FARC presence and zero for municipalities with historical ELN presence. Municipalities assigned to neither historical armed group are excluded from the second design. The resulting conflict comparison contains 216 FARC municipalities and 41 ELN municipalities.

Table 1 reports pre-period characteristics for 2008 to 2011. PDET municipalities are far poorer and have much lower initial achievement than non-PDET municipalities. Ninety-one percent of PDET students are in household strata 1 or 2, compared with 73 percent outside the PDET. Baseline mathematics and verbal achievement are roughly 0.28 standard deviations lower. Near-completion within six years is 6.6 percentage points lower. These differences do not invalidate a fixed effects event study if the groups would otherwise follow parallel trends, but they make extrapolation demanding and rule out a simple cross-sectional interpretation.

The FARC and ELN groups are much closer on educational achievement. Their baseline mathematics difference is 0.03 standard deviations and the verbal difference is effectively zero. Near-completion differs by only 0.62 percentage points, and conditional time to Saber Pro is almost identical. Low socioeconomic status remains more common in FARC municipalities, with a standardized difference of 0.19. Overall, the conflict comparison sacrifices power for greater baseline overlap.

Table 1: Baseline characteristics, 2008 to 2011

	PDET design			Conflict design		
Characteristic	Non-PDET	PDET	Std. diff.	ELN	FARC	Std. diff.
Female, percent	54.73	55.75	0.021	56.35	55.18	−0.023
Low socioeconomic status, percent	73.06	91.16	0.486	85.58	91.63	0.191
Saber 11 mathematics, SD	0.031	−0.253	−0.294	−0.131	−0.161	−0.031
Saber 11 verbal, SD	0.031	−0.250	−0.288	−0.150	−0.153	−0.003
Saber Pro within six years, percent	23.96	17.38	−0.163	17.97	18.59	0.016
Years to Saber Pro, conditional	6.621	6.884	0.089	6.749	6.754	0.002

Notes: Low socioeconomic status identifies household strata 1 and 2. Standardized differences divide the difference in group means by the pooled standard deviation. The conflict design compares municipalities with historical FARC and ELN presence.

4.3 Municipal conflict and uncertainty measures

I organize ten administrative event files into a balanced municipality-year panel for 2007 to 2018 and merge them with CEDE population estimates. The core conflict measure sums subversive actions, attacks on bridges and roads, kidnappings, terrorism, and oil-pipeline bombings. These series cover the full analysis window. Each count is converted to events per 100,000 residents, transformed using $\log(1 + x)$, standardized over the municipality-year panel, and averaged into a core conflict index. Homicide and extortion are analyzed separately because they capture broader public safety and can reflect criminal activity beyond the FARC conflict. Attacks on the security forces begin only in 2010 and are retained as a supplementary outcome. Cocaine records describe seizures or enforcement activity, not coca cultivation, and are not used as a main conflict outcome.

The conflict panel contains 1,122 municipalities and 13,461 municipality-year observations. It includes all 170 PDET municipalities observed in the population data and reproduces the 216 historical FARC and 41 historical ELN municipalities used in the student analysis. Missing municipality-year events are coded as zero only within the observed coverage period of each administrative source. Conflict outcomes are analyzed at the municipality-year level with municipality and year fixed effects, population-normalized dependent variables, and municipality-clustered standard errors.

The uncertainty analysis starts from the same linked student sample and separates observed variables from proxies. I directly observe elapsed years from Saber 11 to the first linked Saber Pro record, the semester reported at Saber Pro, whether tuition is financed with credit or a scholarship, and whether the examinee worked during the previous week. I do not observe applications, the date of initial higher-education enrollment, or semester-by-semester enrollment histories. Consequently, reaching Saber Pro within four or five years measures rapid progression to near-completion, not first-year enrollment. I also construct excess elapsed time as years since Saber 11 minus one-half of the semester reported at Saber Pro. This quantity can reflect delayed entry, interruptions, program changes, irregular semester reporting, or other timing differences. A value of at least one year defines a deliberately labeled delay or interruption proxy.

Financing, employment, semester, and excess-time outcomes are available only among students observed in university-level Saber Pro within six years. They are therefore conditional composition outcomes. The models include Saber Pro examination-period fixed effects when variable definitions or reporting can vary across exam administrations. Desired-institution questions in Saber 11 are not used as application measures because they disappear after the 2014 redesign and record preferences rather than submitted applications.

4.4 Secondary achievement and value added

The secondary school analysis aggregates Saber 11 records to municipality by year. Outcomes include test takers per thousand residents, schools represented per 100,000 residents, cohort standardized mathematics and verbal scores, and student composition. The main pooled specification focuses on the redesigned test era and defines the first fully post-agreement cohort as 2017. Because the test scale changed, scores are standardized within examination cohort before aggregation.

For linked students who appear in Saber Pro, I also construct quantitative and reading value added. Saber Pro scores are standardized within test year and residualized on the corresponding Saber 11 score and baseline covariates. The municipality level residual measures performance near the end of university relative to what would be predicted from secondary school achievement. This is a selected-completer measure: peace can change who reaches Saber Pro, so it is not an unconditional effect on the original Saber 11 cohort.

Figure 2 previews the empirical argument as a sequence of educational margins. The purpose is not to place differently scaled estimates on a common metric, but to make clear that the paper studies participation, progression, speed, learning, and the type of university trajectory rather than a single completion proxy.

5 Empirical Strategy

The main event study is estimated at the student level:

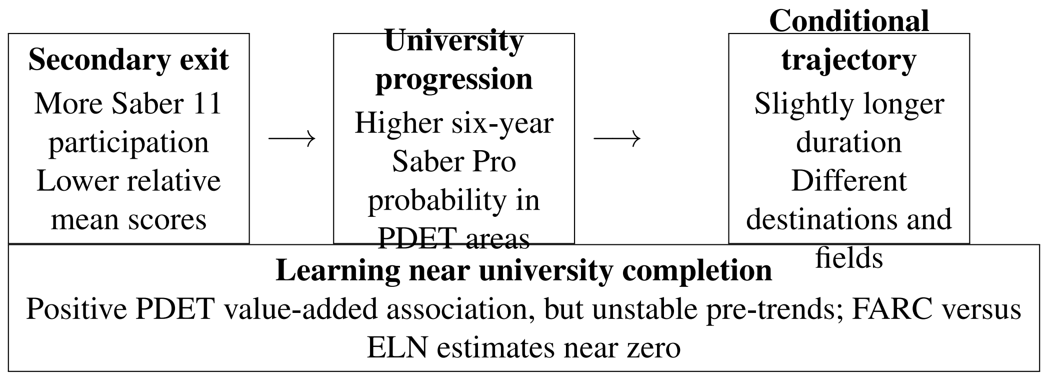


Figure 2: The educational pipeline studied in the paper

Notes: The diagram summarizes the empirical margins and their qualitative direction. Exact estimates and identification diagnostics are reported in Section 5.

$$Y_{imt} = \alpha_m + \lambda_t + \sum_{k \neq 2011} \beta_k \left(\mathbb{1}\{t = k\} \times T_m \right) + \varepsilon_{imt}, \tag{1}$$

where Y_{imt} is an outcome for student i from municipality m in Saber 11 cohort t, α_m are municipality fixed effects, λ_t are cohort fixed effects, and T_m is either PDET status or historical FARC presence. The omitted cohort is 2011, the year before public negotiations. Standard errors are clustered by municipality. For binary outcomes, coefficients are reported in percentage points.

The event coefficients serve two purposes. Pre-2012 estimates provide a direct diagnostic for differential trends. Post-2012 estimates reveal whether adjustment occurred at negotiation, ceasefire, agreement, or implementation. A conventional pooled difference in differences model complements the event study:

$$Y_{imt} = \alpha_m + \lambda_t + \delta \left(T_m \times Post_t \right) + \varepsilon_{imt}. \tag{2}$$

I report pooled effects using 2012 and 2015 as alternative post-period starting dates. The first captures the full negotiation period. The second begins after the December 2014 ceasefire and is closer to a de-escalation effect. For the secondary school municipality panel, I estimate analogous models with municipality and year fixed effects and municipality clustered standard errors.

For the conflict first stage, the unit is municipality-year rather than student. I estimate Equation 1 after replacing the individual outcome with the conflict index or a population-normalized event rate. The main FARC versus ELN model compares 2012 to 2018 with 2007 to 2011 and omits 2011 in the event study. I also report a post-agreement specification beginning in 2017, the first full calendar year after the final agreement. Equal municipality weights are primary, with population-weighted estimates as a robustness check. The conflict index is preferred to any single event category because it reduces outcome selection and averages several administrative measures with common coverage.

The uncertainty models retain 2011 as the reference cohort and estimate three cumulative timing outcomes: Saber Pro within four, five, and six years. These horizons ask whether the shape of progression shifted, not when students first enrolled. Conditional models for elapsed time, semester-based excess time, credit, scholarship, and work include Saber Pro period fixed effects. The uncertainty hypothesis receives its strongest support if the early timing coefficients fall after 2012 and recover after credible de-escalation, while direct proxies move in the predicted direction with stable pre-trends. Because applications and enrollment histories are unobserved, even that pattern would remain suggestive rather than definitive.

The identifying assumption differs across designs. In the PDET comparison, it requires that prioritized and non-prioritized municipalities would have experienced similar changes after 2011 absent the peace process and its territorial consequences. This assumption is plausible for the near-completion outcome based on the available pre-coefficients, but less plausible for several score outcomes that exhibit differential pre-trends. The PDET estimates are therefore presented as a disciplined event study of prioritized territories, with causal language reserved.

In the FARC versus ELN design, the assumption is that education in historically ELN municipalities approximates the counterfactual trajectory of historically FARC municipalities after common national shocks. This design removes many differences between conflict and non-conflict Colombia and matches the treatment contrast in Fajardo-Steinhäuser (2024). It can still fail if FARC and ELN territories faced different post-2012 shocks, if armed groups selected systematically different locations, or if spillovers affected ELN areas. With only 41 ELN municipalities, confidence intervals are also wider than in the PDET analysis.

No single comparison is decisive. The research design is intentionally triangulated. A result is most credible when it appears in both designs, follows the timing of de-escalation, has stable pre-trends, and is consistent across related outcomes. Disagreement between designs is itself informative about external validity and mechanisms.

6 Results

6.1 Conflict as the first stage

Figure 3 begins with the outcome closest to the treatment itself. Historical FARC municipalities experience a pronounced decline in the standardized core conflict index relative to historical ELN municipalities after negotiations begin. The pooled post-2012 coefficient is -0.422 with a standard error of 0.094. For the log core-event rate, the corresponding estimate is -0.656 with a standard error of 0.129. Joint tests do not reject the 2007 to 2010 pre-treatment coefficients, with p values of 0.462 for the index and 0.491 for the log event rate. The negative differential appears in 2012, becomes larger in 2013, and persists through 2018.

The PDET pattern is different. There is little pooled change during the negotiation period, followed by a large relative decline after the agreement. The post-2017 conflict-index coefficient is -0.423 with a standard error of 0.050, and the log event-rate estimate is -0.524 with a standard error of 0.072. However, PDET pre-trends are jointly significant for both measures. This is useful descriptive evidence that conflict fell sharply in prioritized territories after 2016, but it is not as credible a causal contrast as FARC versus ELN.

Broader public-safety outcomes do not move uniformly with guerrilla-related violence. In PDET municipalities, reported extortion rises relative to non-PDET municipalities during both the negotiation and post-agreement periods, and homicide rises after the agreement. In FARC municipalities, homicide falls relative to ELN municipalities during negotiations, while extortion is statistically indistinguishable from zero. This divergence cautions against describing demobilization as the disappearance of all insecurity. The core conflict reduction can coexist with criminal substitution, reporting changes, or entry by other armed actors. Appendix Figure 13 reports the pooled estimates across outcomes.

The first stage changes how the education results should be read. The FARC versus ELN education

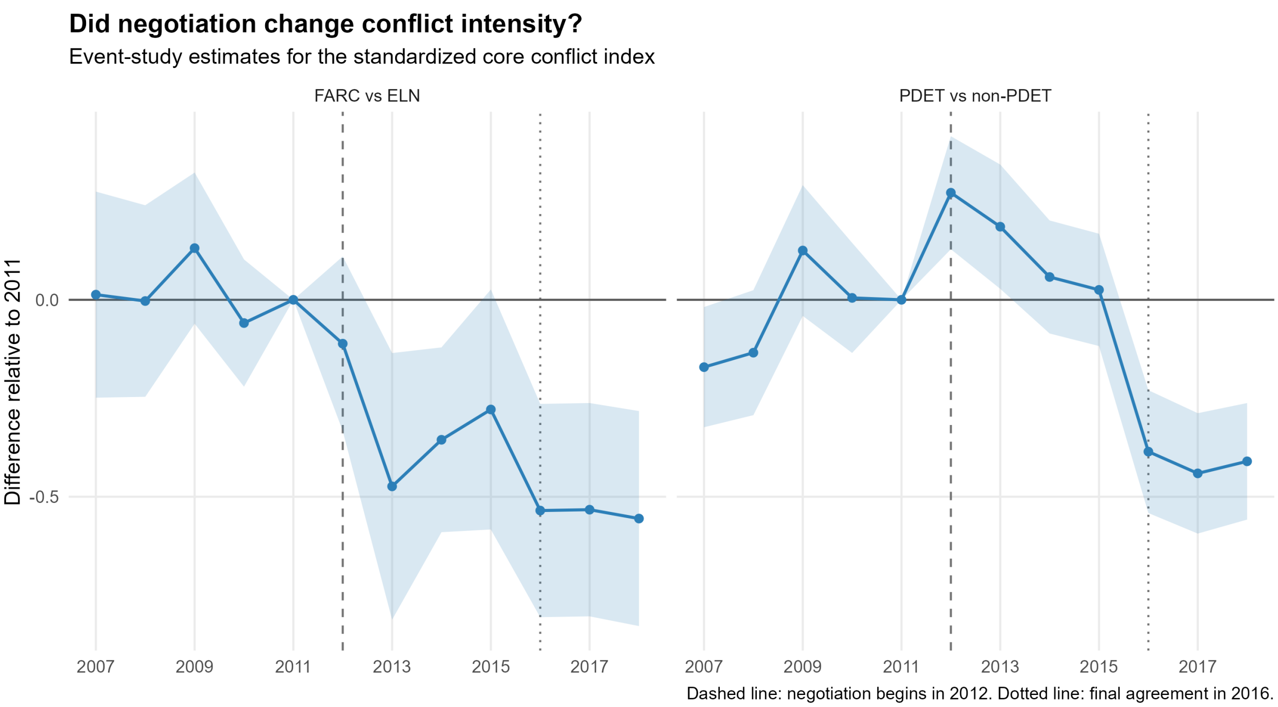


Figure 3: Conflict intensity in historical FARC and ELN municipalities and PDET territories

Notes: The outcome is the standardized core conflict index constructed from subversive actions, attacks on bridges and roads, kidnapping, terrorism, and oil-pipeline bombing. Models include municipality and year fixed effects and omit 2011. The left panel compares historical FARC with historical ELN municipalities. The right panel compares PDET with non-PDET municipalities. Bands are 95 percent confidence intervals from municipality-clustered standard errors. Dashed and dotted lines denote the beginning of negotiations and the final agreement.

null does not arise because the treatment failed to alter conflict. On the administrative outcomes observed here, conflict fell substantially and persistently in FARC territories, yet the education pipeline did not differentially improve. This pattern closely matches the state-capacity interpretation in Fajardo-Steinhäuser (2024): security is necessary for a peace dividend, but it is not sufficient when complementary public and market institutions remain weak.

6.2 Progression to the end of university

Figure 4 presents the central PDET event study. The coefficients before 2012 are small and jointly insignificant, with a pre-trend test p value of 0.274. At the start of negotiations, the relative probability that a PDET student reaches Saber Pro within six years falls by 1.17 percentage points. The 2013 estimate remains negative but is less precise. A break occurs in 2014: the coefficient rises to 1.74 percentage points and remains positive thereafter. It reaches 2.50 points in 2016 and 3.15 points in 2018.

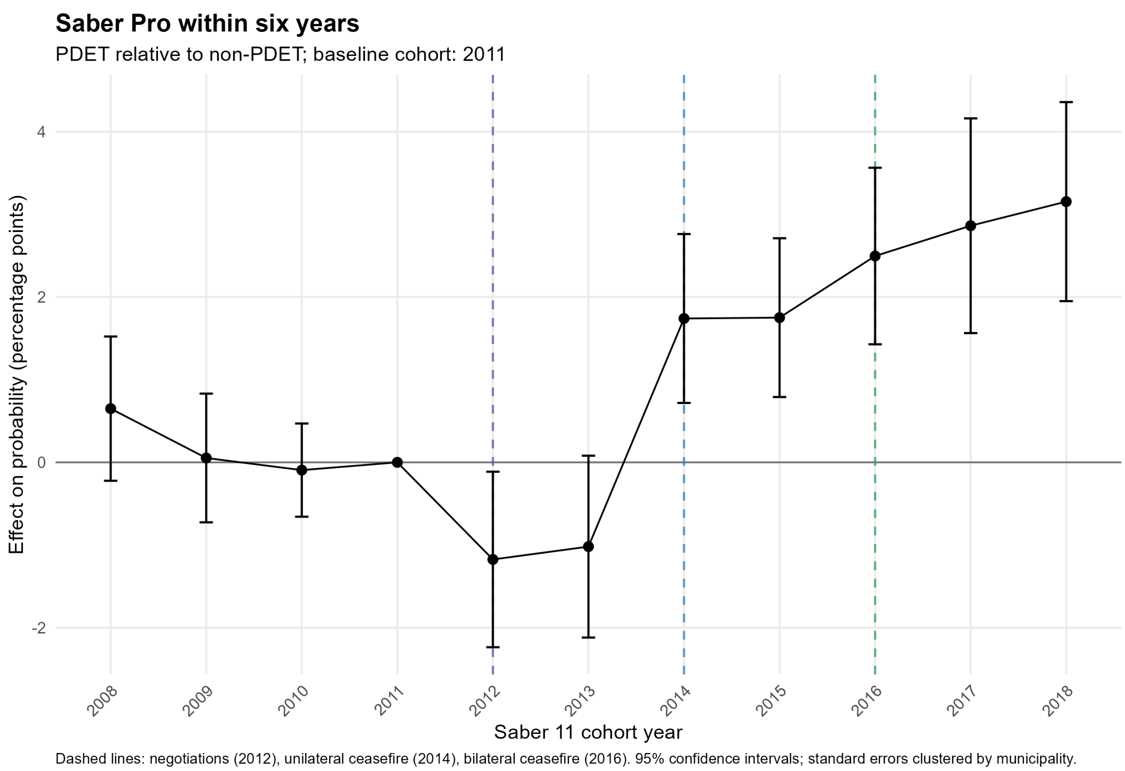


Figure 4: PDET event study: Saber Pro within six years

Notes: Coefficients are interactions between cohort indicators and PDET status in a model with municipality and cohort fixed effects. The omitted cohort is 2011. Bars are 95 percent confidence intervals from municipality clustered standard errors. Estimates are percentage points.

The timing is substantively revealing. If negotiation alone changed educational expectations, an improvement should begin in 2012. Instead, the initial coefficient is negative and the reversal coincides with the ceasefire period. A credible reduction in violence, rather than the political announcement of talks, appears to be the relevant turning point. The result resembles evidence that human capital and entrepreneurship responded when the ceasefire reduced violence and uncertainty (Namen et al., 2021; Bernal et al., 2024).

The magnitude is meaningful. In the PDET pre-period, 17.4 percent of students reach Saber Pro within six years. A 3.15 percentage point increase is therefore about 18 percent of that baseline rate. Yet this is a relative effect. It does not mean that the raw completion share rose by three points, and it does not establish that PDET designation itself caused the change. National cohort trends are negative over this period, as Figure 1 shows. The estimate says that prioritized municipalities improved relative to the counterfactual movement of non-PDET municipalities after fixed differences and national cohort shocks are removed.

Figure 5 subjects the same outcome to the FARC versus ELN design. The post-period coefficients fluctuate near zero and every confidence interval includes zero. The pooled post-2012 estimate is 0.11 percentage points with a standard error of 0.81, and the pooled post-2015 estimate is 0.34 points with a standard error of 0.74. Pre-treatment coefficients are jointly insignificant with a p value of 0.329. Thus, the clean conflict comparison provides no evidence that students from FARC municipalities differentially reached Saber Pro after the ceasefire.

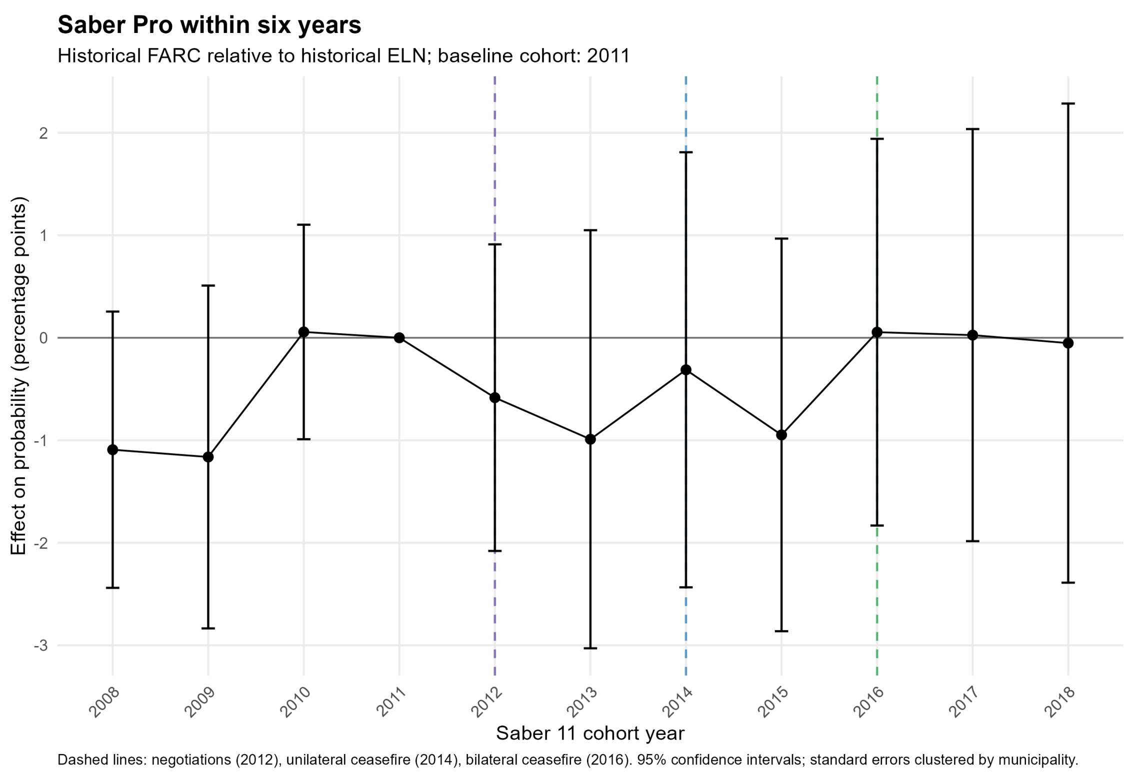
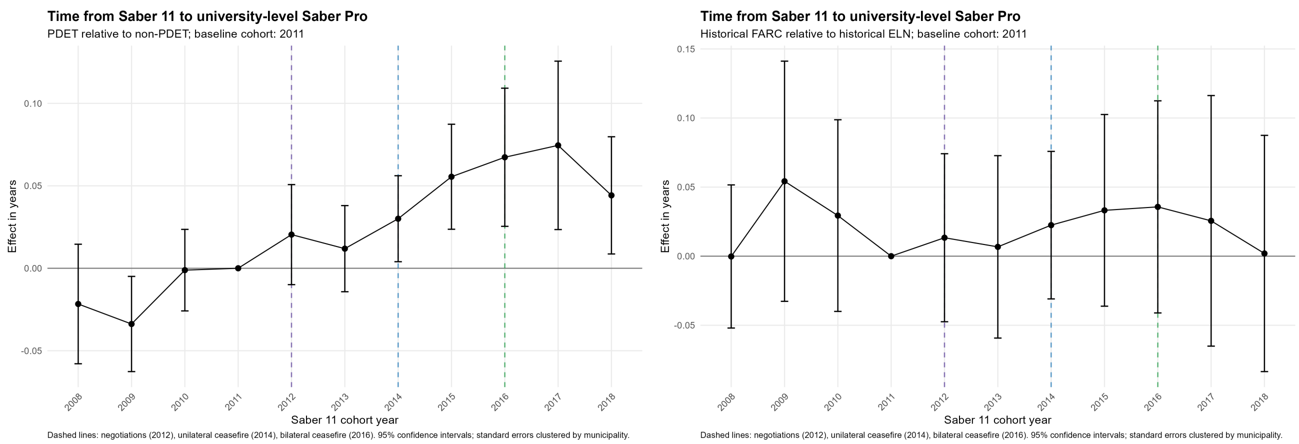


Figure 5: FARC versus ELN event study: Saber Pro within six years
Notes: Coefficients compare historical FARC municipalities with historical ELN municipalities. The omitted cohort is 2011. The model includes municipality and cohort fixed effects. Bars are 95 percent confidence intervals from municipality clustered standard errors. Estimates are percentage points.

The contrast between Figures 4 and 5 is the paper's most important inferential fact. The PDET pattern should not be generalized into a universal effect of FARC demobilization. One interpretation is that PDET areas combined security changes with anticipation or delivery of territorially targeted policy. Another is that the broad non-PDET comparison captures differential recovery unrelated to peace. A third is statistical: the ELN sample is small and its estimates are less precise. Still, the pooled FARC versus ELN interval rules out very large average effects and centers close to zero.

6.3 Speed of progression among completers

If peace improved the efficiency of the education pipeline, students who reached Saber Pro might do so faster. The data do not support that mechanism. In the PDET design, conditional time to Saber Pro begins to rise in 2014. The estimate is 0.030 years in 2014, 0.056 in 2015, 0.067 in 2016, and 0.075 in 2017. The 2017 increase is about 0.9 months. The pre-trend test is not significant, with a p value of 0.144.



(a) PDET relative to non-PDET

(b) Historical FARC relative to historical ELN

Figure 6: Time from Saber 11 to university-level Saber Pro

Notes: The sample is conditional on appearing in university-level Saber Pro. Coefficients are measured in years relative to the omitted 2011 cohort. The PDET and FARC versus ELN pre-trend p values are 0.144 and 0.553, respectively.

This result is not contradictory to higher near-completion. It suggests that the extensive margin and the intensive margin moved differently. Newly progressing students may be more likely to interrupt study, enroll part time, attend institutions farther away, or begin with weaker preparation. Bringing those students near completion within six years raises access while increasing average duration among completers. The result also warns against conditioning mechanically on Saber Pro participation. As peace changes selection into the observed completion sample, conditional outcomes can worsen even when the unconditional probability of reaching that sample improves.

6.4 Uncertainty, timing, and financing

Figure 7 tests whether the initial decline is specific to the six-year endpoint. It is not. For PDET-origin students, the 2012 coefficient is -1.00 percentage points for Saber Pro within four years, -1.23 points within five years, and -1.17 points within six years. All three series remain negative in 2013, reverse sharply in 2014, and increase thereafter. By 2018, the corresponding estimates are 2.15, 2.98, and 3.15 percentage points. This common shape suggests a temporary delay in the pipeline rather than an artifact of choosing a six-year threshold.

The credibility of the three horizons is not identical. The six-year PDET outcome has a pre-trend p value of 0.274, while the four- and five-year outcomes reject parallel pre-trends. I therefore treat the repeated post-2012 shape as corroborating timing evidence, not three independent causal estimates. The FARC versus ELN timing paths remain statistically indistinguishable from zero at all horizons and have

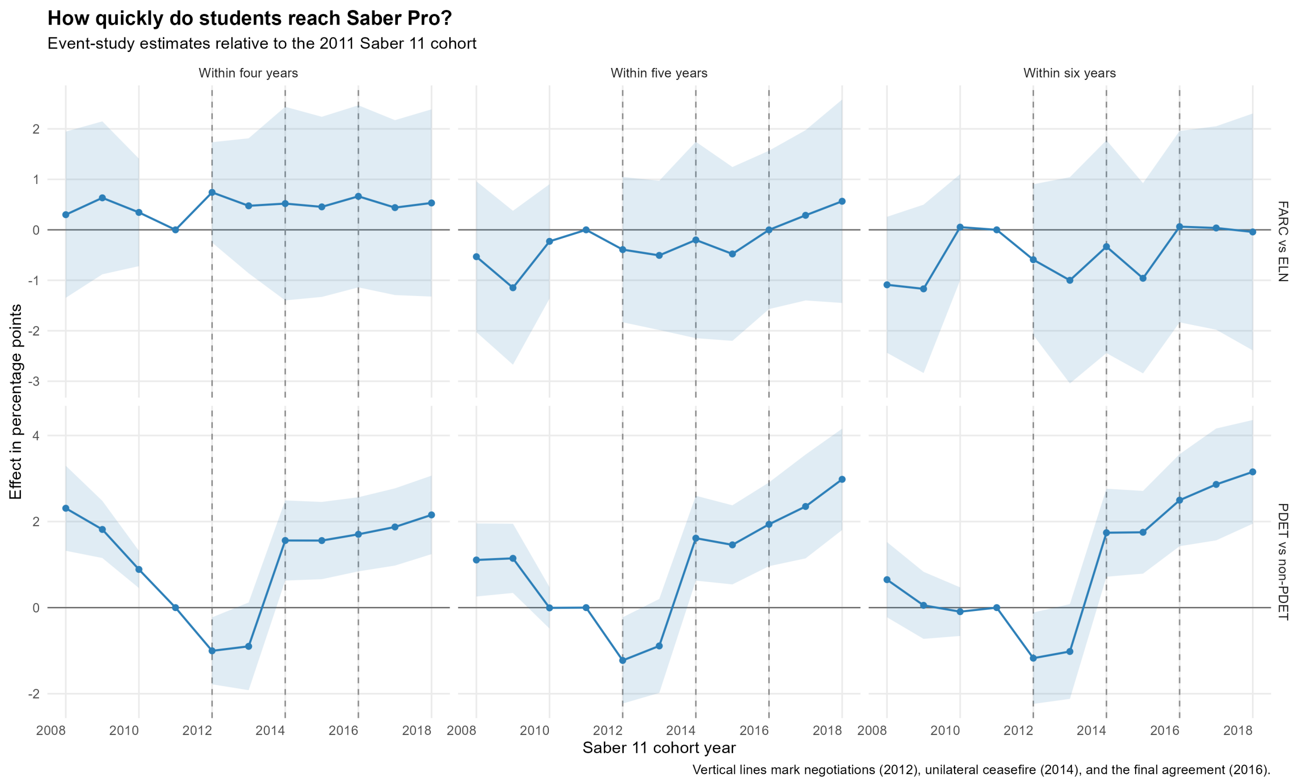


Figure 7: Timing of progression to Saber Pro

Notes: Panels report event-study coefficients for appearing in Saber Pro within four, five, and six years of Saber 11. The omitted cohort is 2011. The top row compares historical FARC with historical ELN municipalities; the bottom row compares PDET with non-PDET municipalities. Estimates are percentage points with 95 percent confidence intervals from municipality-clustered standard errors. These are near-completion horizons, not first-enrollment outcomes.

acceptable pre-trends. Thus, the uncertainty pattern is specific to the PDET comparison and does not explain the cleaner FARC versus ELN education null.

Direct proxies provide little support for a financing or interruption channel. In pooled PDET models, credit financing changes by 0.28 percentage points with a standard error of 1.21, while the delay or interruption proxy changes by -0.61 points with a standard error of 0.82. Excess elapsed time relative to reported semester falls by 0.020 years and is not statistically significant. The observed time from Saber 11 to university-level Saber Pro nevertheless rises by 0.057 years, about 0.68 months, with stable pre-trends. The difference is informative: completers take slightly longer to reach Saber Pro, but reported semester does not show that the extra time reflects a clean increase in interruption or delayed entry.

The FARC versus ELN mechanism estimates are also mostly null. Credit use changes by 0.63 percentage points with a standard error of 2.12, the delay or interruption proxy changes by 1.16 points with a standard error of 1.49, and total elapsed time is effectively unchanged. Scholarship financing falls by 3.90 percentage points among FARC-origin university examinees, with acceptable pre-trends, but this conditional result can reflect changes in who reaches Saber Pro, institution choice, or national aid programs. Appendix Figures 14 and 15 report these estimates.

These results support a bounded interpretation. The initial PDET decline is consistent with uncertainty during negotiations, because it appears across completion horizons and disappears when de-escalation becomes credible. The administrative data do not observe applications or entry dates, however, and the available financing and semester proxies do not move as a strong uncertainty mechanism would predict. The paper therefore uses uncertainty to interpret the timing, not to claim a separately identified causal channel.

6.5 Secondary school participation and achievement

Figure 8 reports pooled secondary school effects for the redesigned examination era. PDET municipalities have 0.69 additional Saber 11 test takers per thousand residents relative to non-PDET municipalities after the agreement. They also have 3.34 additional represented schools per 100,000 residents, although this estimate is only marginally precise. At the same time, mean mathematics falls by 0.055 standard deviations and mean verbal performance falls by 0.031 standard deviations.

This pattern is naturally read as composition, not proof of declining school quality. When more disadvantaged or rural students remain in school long enough to take Saber 11, the mean score of examinees can fall even if no individual learns less. In fact, the PDET post-period is associated with a lower measured low socioeconomic share but a higher rural school share, suggesting multiple compositional forces. The administrative exam data do not observe students who never reach Saber 11, so they cannot separate a change in individual learning from a change in who is tested.

The FARC versus ELN estimates again center close to zero. Test taking rises by 0.42 per thousand residents, but the standard error is 0.31 and the estimate is not statistically significant. Mathematics changes by -0.004 standard deviations and verbal performance by -0.014, both imprecise. These results reinforce the conclusion that the FARC peace process did not generate a general achievement dividend relative to other conflict-affected municipalities.

6.6 Learning between Saber 11 and Saber Pro

The linked examinations make it possible to ask whether students improved their relative performance between the end of secondary school and the end of university. Figure 9 compares the PDET and FARC

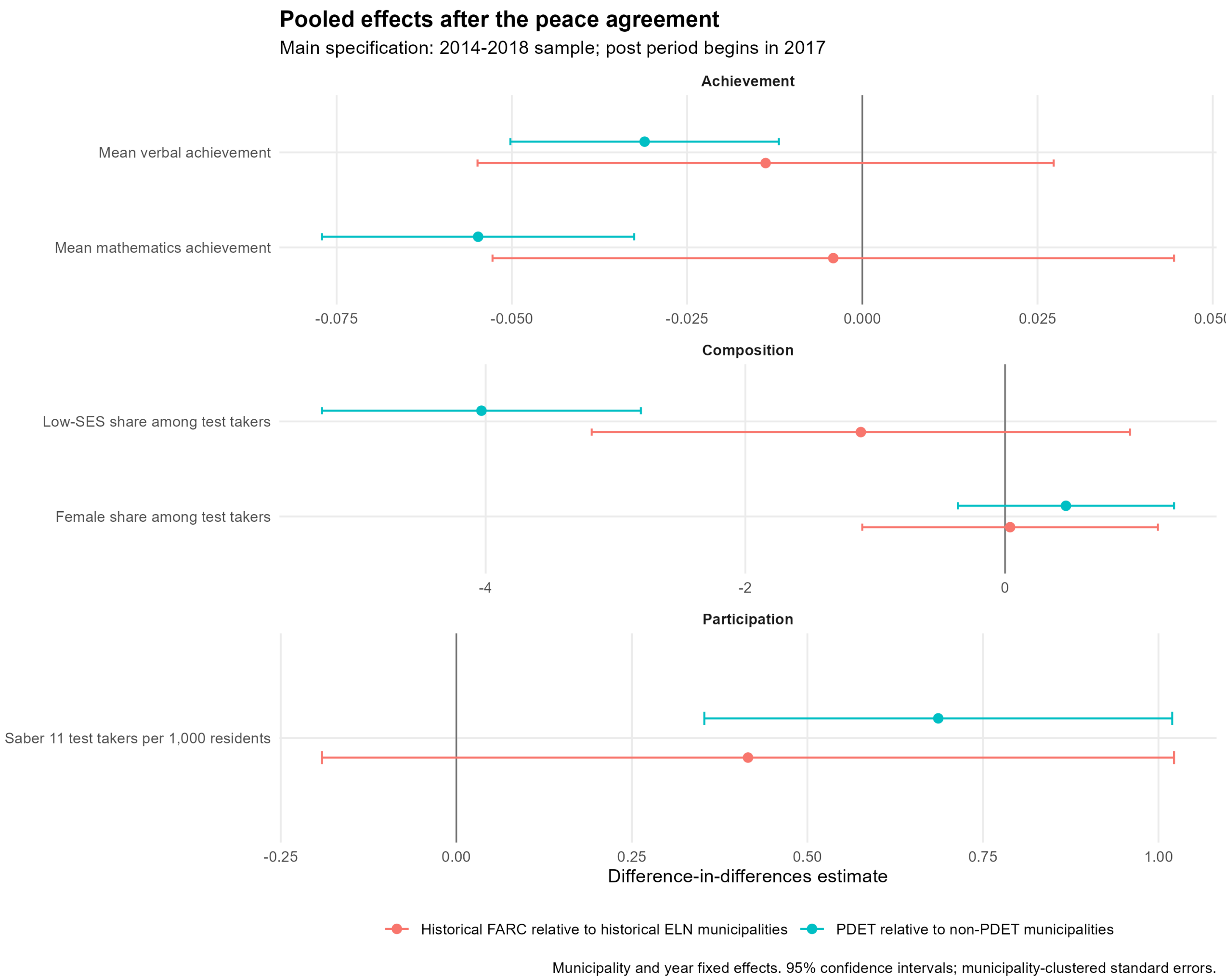
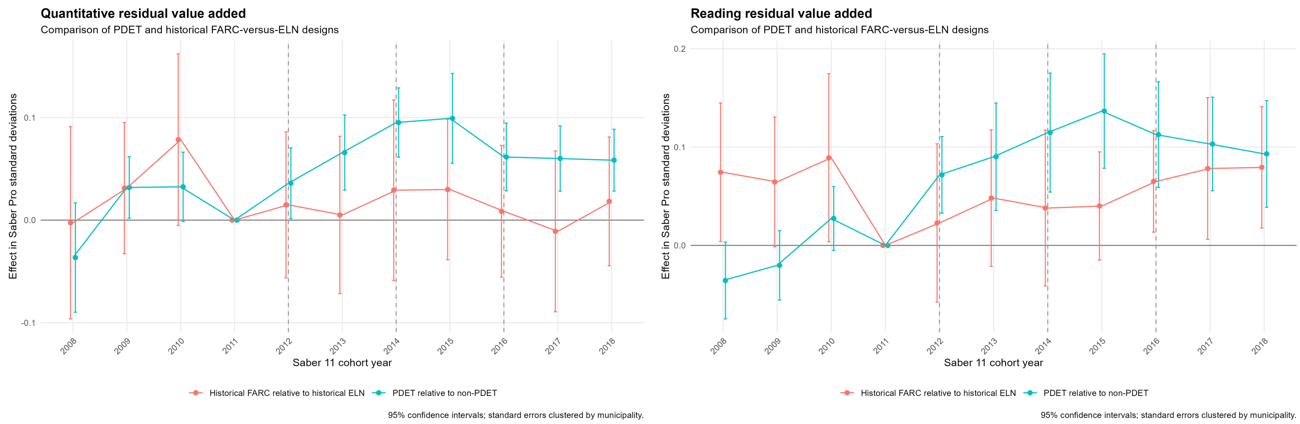


Figure 8: Pooled post-agreement effects on Saber 11 participation and achievement

Notes: Points report treatment by post-agreement coefficients from municipality level models with municipality and year fixed effects. The main specification uses the redesigned Saber 11 era and treats 2017 as the first fully post-agreement cohort. Scores are standardized within cohort. Standard errors are clustered by municipality.

versus ELN event studies for residual quantitative and reading value added. In pooled PDET regressions, the post-2012 associations are 0.060 standard deviations for quantitative value added and 0.109 standard deviations for reading. The event paths are positive from 2012 onward and appear economically meaningful.



(a) Quantitative residual value added

(b) Reading residual value added

Figure 9: Value added between Saber 11 and Saber Pro

Notes: Saber Pro scores are standardized within test year and residualized on the corresponding Saber 11 score and baseline covariates. Estimates condition on being matched to Saber Pro. The PDET pre-trend p values are below 0.01 for both outcomes. The FARC versus ELN pre-trend p values are 0.131 for quantitative and 0.063 for reading value added.

These PDET estimates are suggestive rather than causal for two reasons. First, their event studies reject stable pre-trends: the quantitative and reading pre-trend p values are approximately 0.00003 and 0.009. Second, value added is observed only for students who reach Saber Pro, and the main progression result shows that peace may change precisely who enters that selected group. Positive residual gains can therefore combine institutional learning with changes in selection.

The FARC versus ELN design is cleaner and yields a null. Post-2012 quantitative value added is -0.014 standard deviations with a standard error of 0.019, while reading value added is -0.002 with a standard error of 0.023. These estimates are small relative to the PDET associations and their confidence intervals include zero. The evidence does not support a general improvement in learning caused by the FARC ceasefire.

6.7 University destination, migration, and field of study

The type of university trajectory also matters. Because unconditional university-specific outcomes often have differential PDET pre-trends, I emphasize composition among students who are observed in university-level Saber Pro. These estimates answer whether the destination of completers changed, not whether the original cohort became more likely to complete a particular type of program. Figure 11 and Figure 12 in the appendix report the full event paths.

The migration results do not reveal a robust mechanism. Among PDET completers, the probability of studying outside the home municipality is generally one to two percentage points lower after negotiations, but confidence intervals usually include zero. In the FARC versus ELN comparison, the point estimates are positive, ranging from about 3.7 to 7.6 percentage points between 2012 and 2016, but the intervals widen sharply because the ELN comparison group is small. Peace did not clearly eliminate the need to leave the home municipality, nor is there precise evidence that it increased educational migration.

The most coherent composition result concerns public universities. Among FARC-origin completers, the public-university share is approximately 4.5 to 7.2 percentage points lower than among ELN-origin completers throughout the post-2012 period. The pre-trend test does not reject at conventional levels, with a p value of 0.157. This pattern could reflect expansion through private institutions, geographic differences in public supply, or selection among the students who reach Saber Pro. It should not be read as evidence that public universities contracted because of peace, but it does show that the margin of adjustment was not an across-the-board expansion of public provision.

There is no robust movement into STEM. PDET estimates become modestly negative after 2015, while FARC versus ELN estimates fluctuate around zero with wide confidence intervals. Education programs show a more notable decline among PDET completers, reaching -2.67 percentage points in 2018 with a standard error of 1.20 and acceptable pre-trends. FARC versus ELN estimates for education are also generally negative after 2014, although less precise. Together, these results suggest that expanded progression did not concentrate in the fields most directly associated with local public-service provision or technological upgrading.

Table 2 collects the main educational estimates. The appendix reports the conflict first stage and uncertainty regressions separately so their samples and scales remain transparent. The tables distinguish visually strong PDET dynamics, conditional outcomes, and FARC versus ELN nulls rather than treating statistical significance as a substitute for design credibility.

7 Interpretation: An Access Dividend with Capacity Constraints

The results are best understood as a sequence from security to educational capacity. In the FARC versus ELN comparison, the first link is strong: core conflict falls substantially after negotiations begin and the pre-period paths are stable. The next links are weak. Secondary participation, scores, progression to Saber Pro, and university value added do not differentially improve. This combination is more informative than either result alone. It shows that a real peace dividend in security need not become a human-capital dividend within the observed horizon.

The PDET estimates instead reveal movement at the access margin. Educational progression responds after de-escalation becomes credible, and the simultaneous rise in Saber 11 participation and near-completion is consistent with a wider pipeline. The longer time to Saber Pro and the absence of robust score or value-added improvements imply that the pipeline did not become uniformly more productive. Since PDET status also captures subsequent territorial attention and policy, this pattern can coexist with the FARC versus ELN null if access requires complements beyond reduced violence.

Several mechanisms can generate this combination. Lower violence can reduce displacement, recruitment risk, road insecurity, and the direct cost of attending school. Families may also lengthen their planning horizon once a ceasefire appears durable. These mechanisms primarily affect whether a student participates. Learning and speed, by contrast, depend on teachers, schools, university places, transport, finance, connectivity, and institutional quality. Such inputs adjust slowly, especially in municipalities selected precisely because poverty and state weakness are severe.

The findings therefore fit a state-capacity interpretation of the peace dividend. Fajardo-Steinhäuser (2024) finds that violence fell sharply in historical FARC areas without a parallel improvement in broad economic activity, and links the null to weak state entry. This paper reproduces the sharp relative conflict

Table 2: Selected estimates

Outcome and comparison	Estimate	Std. error	95 percent interval
Panel A: PDET relative to non-PDET municipalities			
Saber Pro within six years, 2012 event coefficient, pp	−1.175	0.542	[−2.236, −0.113]
Saber Pro within six years, 2014 event coefficient, pp	1.740	0.521	[0.719, 2.762]
Saber Pro within six years, 2016 event coefficient, pp	2.496	0.545	[1.428, 3.564]
Saber Pro within six years, 2018 event coefficient, pp	3.154	0.614	[1.950, 4.358]
Years from Saber 11 to Saber Pro, 2017, years	0.075	0.026	[0.023, 0.126]
Saber 11 test takers per 1,000 residents, post 2017	0.686	0.170	[0.353, 1.020]
Saber 11 mathematics, post 2017, SD	−0.055	0.011	[−0.077, −0.033]
Saber 11 verbal, post 2017, SD	−0.031	0.010	[−0.050, −0.012]
University quantitative value added, post 2012, SD	0.060	0.017	[0.026, 0.094]
University reading value added, post 2012, SD	0.109	0.024	[0.061, 0.156]
Education field among completers, 2018, pp	−2.669	1.205	[−5.030, −0.308]
Panel B: Historical FARC relative to historical ELN municipalities			
Saber Pro within six years, pooled post 2012, pp	0.112	0.810	[−1.475, 1.699]
Saber Pro within six years, pooled post 2015, pp	0.338	0.744	[−1.121, 1.797]
Saber 11 test takers per 1,000 residents, post 2017	0.415	0.310	[−0.191, 1.022]
Saber 11 mathematics, post 2017, SD	−0.004	0.025	[−0.053, 0.044]
Saber 11 verbal, post 2017, SD	−0.014	0.021	[−0.055, 0.027]
University quantitative value added, post 2012, SD	−0.014	0.019	[−0.051, 0.024]
University reading value added, post 2012, SD	−0.002	0.023	[−0.047, 0.043]
Public university among completers, 2018, pp	−4.486	2.456	[−9.301, 0.328]
Studied outside home municipality, 2018, pp	5.480	5.140	[−4.594, 15.553]
STEM field among completers, 2018, pp	1.625	2.547	[−3.366, 6.617]

Notes: Event coefficients use 2011 as the omitted cohort. Pooled estimates include municipality and cohort or year fixed effects. Standard errors are clustered by municipality. "pp" denotes percentage points. Time, value added, destination, and field estimates condition on students observed in Saber Pro. PDET value-added estimates have differential pre-trends and are reported as associations rather than causal effects.

decline using a different administrative event panel and shows the same disconnect for education. Security makes investment possible, but public and private capacity determines whether possibility becomes achievement. The positive but transitory firm entry documented by Bernal et al. (2024) similarly shows that expectations and complementary programs mediate the response to de-escalation.

Uncertainty remains a plausible explanation for why PDET progression falls at the beginning of negotiations and reverses around the ceasefire. Education is a long-horizon investment, and households facing insecure income, travel, or territorial control may wait for information before committing (Kazianga, 2012; Stange, 2012). Yet the mechanism evidence is deliberately limited. Credit use and the semester-based delay or interruption proxy do not change detectably, and first enrollment is unobserved. The safer conclusion is that the dynamic path is consistent with uncertainty, while the data do not identify the household decision that generated it.

The quantity and quality divergence also connects this paper to existing education studies. Namen et al. (2021) document improvements in several educational outcomes after the ceasefire. Montaño-Bardales and Palacios (2025) find greater access after the 2016 agreement but wider relative gaps in dropout and standardized performance. Avendaño Santiago (2021) emphasizes student recomposition as a mechanism for lower observed Saber 11 performance. The present results extend that logic: bringing additional students through secondary school and toward university can lower mean scores and lengthen duration even while expanding opportunity.

The null FARC versus ELN results should not be hidden behind the PDET estimates. They discipline the story in two ways. First, they show that removing FARC violence alone was not sufficient to produce a detectable average change in the education pipeline relative to ELN territories. Second, they suggest that any PDET access dividend may depend on targeting, expectations, or characteristics of the prioritized areas rather than the peace agreement in isolation. A null is substantively informative when theory predicts that security and state capacity are complements.

The destination and field results sharpen this interpretation. The expansion is not accompanied by a robust movement into STEM, and education programs become relatively less common among PDET-origin completers. In the FARC versus ELN design, the persistent decline in public-university participation among completers suggests that any adjustment may have depended on private institutions or on leaving the public sector unchanged while student demand evolved. These are conditional composition patterns, but they are difficult to reconcile with a large, coordinated expansion of public higher-education capacity in former conflict territories.

8 Threats to Identification and Scope

The first limitation is treatment definition. PDET status was established in 2017, after negotiations and the ceasefire. It is a marker for territories prioritized by the peace architecture, not a policy shock that began in 2012. Since designation reflects poverty, rurality, illicit crops, and conflict, differential recovery can arise for reasons other than peace. Municipality and cohort fixed effects remove stable gaps and national shocks, while pre-trend tests support the main near-completion outcome, but neither feature proves the counterfactual.

Second, the FARC versus ELN design has only 41 control municipalities. Its confidence intervals are wider, historical presence can be measured with error, and the ELN is not a perfect untreated group. ELN territories experienced national political change and may have been affected by spillovers, changing armed competition, or their own negotiations. The design estimates a relative effect between two types of conflict exposure, not the effect of peace relative to continued war in a sealed laboratory.

Third, municipality is measured at Saber 11. Students may migrate for university, so the outcome captures whether youth originating in a municipality reach Saber Pro, not whether higher education is locally supplied. That origin-based interpretation is valuable for human capital opportunity but does not identify local university expansion. Future work could use institution and campus identifiers to separate local supply, educational migration, and destination quality.

Fourth, appearing in Saber Pro is not identical to graduating. It indicates completion of at least 75 percent of a professional program and usually occurs close to graduation. Some examinees may not graduate, and graduates of technical or technological programs are excluded because Saber TyT lacks a comparable pre-period. The outcome is therefore a conservative, professional higher education progression measure.

Fifth, the linked data condition on taking Saber 11. If peace primarily changes whether students reach the exam, the postsecondary analysis misses youth who remain outside the observed secondary school population. The municipality level Saber 11 participation results partly address this concern and suggest that selection into the starting sample changed. A complete welfare analysis would link population cohorts, enrollment records, dropout, secondary completion, higher education enrollment, and formal graduation.

Sixth, the conflict panel measures recorded events rather than latent territorial control or victimization. Reporting can change with police presence, access, or institutional capacity. The core index reduces dependence on a single category and the FARC versus ELN pre-trends are stable, but administrative zeros need not mean an absence of insecurity. Homicide and extortion are kept separate precisely because the end of FARC operations can change the composition of violence without ending all criminal activity.

Seventh, the uncertainty tests are constrained by the examination architecture. Saber Pro reveals that a student reached the final portion of a program, not when the student applied or first enrolled. Reported semester is a contemporaneous self-report and can be affected by program changes, transfers, part-time study, and inconsistent semester counting. Financing and employment are observed only among eventual university-level Saber Pro examinees. These outcomes can describe the selected completion path, but they cannot replace administrative enrollment, application, ICETEX, or semester-by-semester persistence records.

Finally, multiple outcomes raise the risk of selective emphasis. The paper gives priority to a pre-specified conceptual hierarchy: the conflict first stage, unconditional progression within a fixed horizon, participation at the secondary margin, conditional speed, and score-based learning. It also foregrounds pre-trend diagnostics and the comparison across designs. Several statistically significant PDET score and short-horizon timing estimates are not treated as causal because their pre-trends are unstable. This conservative reporting is essential in a setting with many plausible outcomes.

9 Conclusion

Colombia's peace process generated a substantial security dividend without a corresponding general educational dividend. Historical FARC municipalities experience a large and persistent decline in core conflict relative to historical ELN municipalities, with stable pre-trends. Nevertheless, the same comparison shows no detectable improvement in Saber 11 participation or achievement, progression to Saber Pro, or university value added. This is the paper's strongest conclusion: reducing organized armed violence was consequential, but insufficient by itself to transform human-capital accumulation.

The education pipeline was not entirely unchanged. Among students originating in municipalities later prioritized by the PDET, relative progression to Saber Pro first deteriorates when negotiations begin and improves sharply after the 2014 ceasefire. By the 2018 secondary-school cohort, the relative increase reaches 3.15 percentage points, and Saber 11 participation also rises. The initial decline appears at four-, five-, and six-year progression horizons, which is consistent with temporary uncertainty. Yet credit financing and the available delay or interruption proxy remain close to zero, and applications and initial enrollment are not observed. Uncertainty is therefore a plausible reading of the timing, not an identified mechanism.

The broader evidence remains sobering. Students who reach Saber Pro take slightly longer rather than less time, achievement gains are not robust, migration and STEM estimates are imprecise, education programs become less common among PDET-origin completers, and the public-university share falls among FARC-origin completers relative to ELN-origin completers. The strongest conclusion is conditional: peace can open an educational door when accompanied by territorial attention, but security alone does not determine what students find behind it.

This distinction has direct policy relevance. In post-conflict regions, security policy and educational policy are complements. Lower violence can increase demand and participation quickly. Converting that response into learning and timely completion requires teachers, transport, finance, local institutions, and credible state presence. Judging peace only by immediate score gains would miss an access response. Judging it only by increased participation would miss the capacity constraints that prevent access from becoming a durable human capital dividend.

A Additional Descriptive Evidence

Figure 10 displays raw standardized achievement by cohort. The persistent PDET gap makes clear why comparisons in levels are uninformative. FARC and ELN trends overlap much more closely, supporting their use as a conflict-exposed comparison while also showing that the design has limited variation from which to estimate large differential changes.

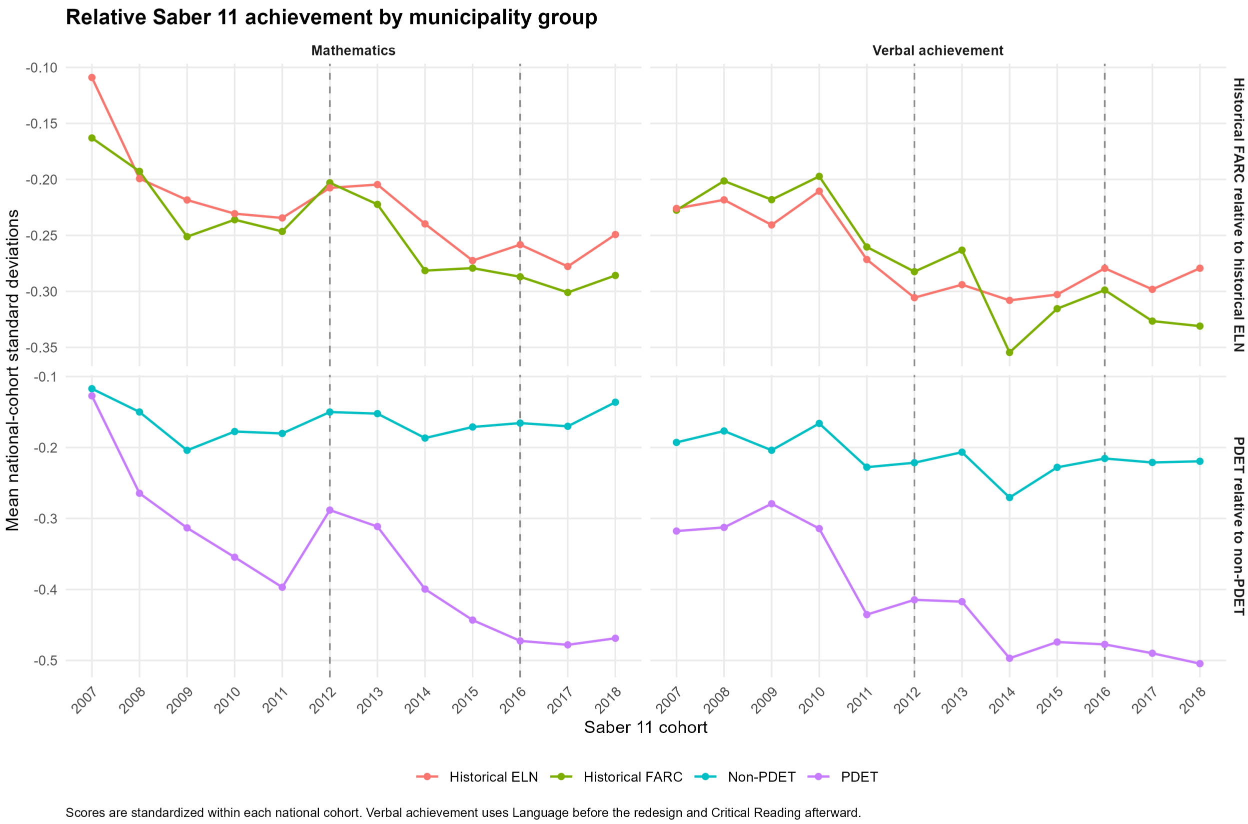
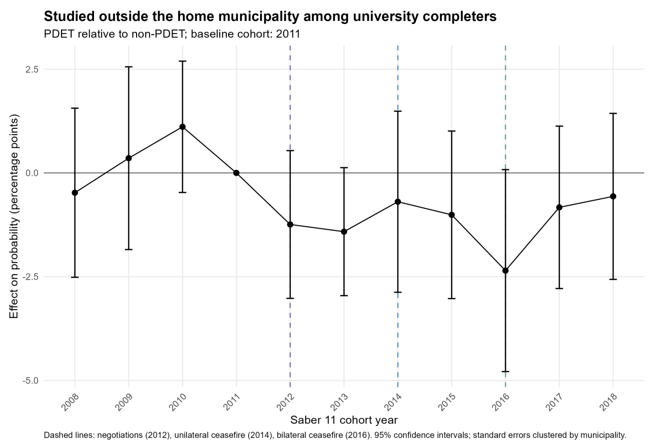


Figure 10: Raw Saber 11 achievement trends

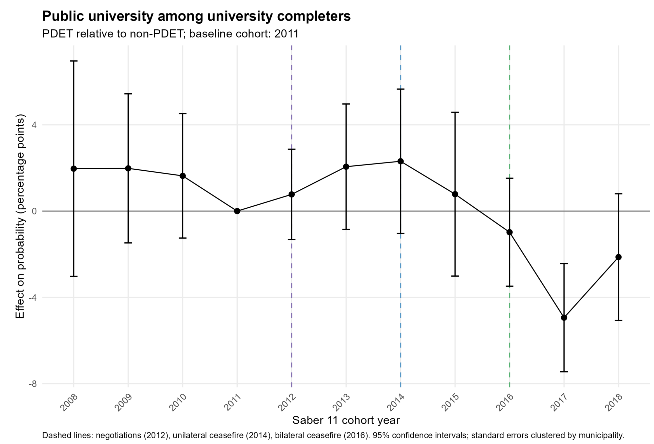
Notes: Mathematics and verbal scores are standardized within test cohort. Lines report group means before regression adjustment.

B Conditional University Trajectories

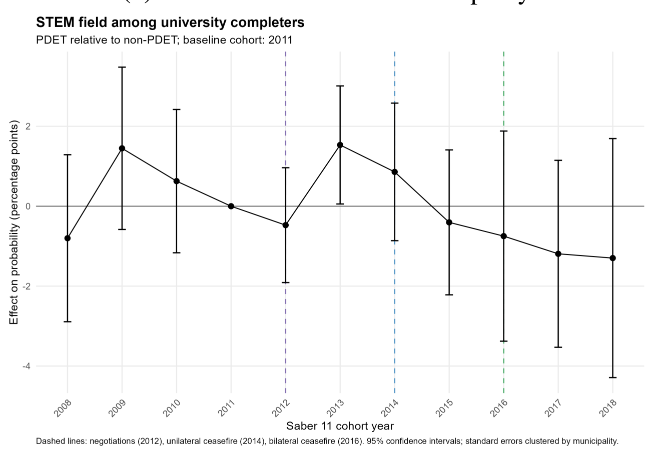
The panels below report outcomes measured among students observed in university-level Saber Pro. Conditioning on university progression makes these composition results potentially endogenous to the peace process. They are included to characterize the students and institutions behind the main extensive-margin result, not as independent causal estimates for the original Saber 11 cohort.



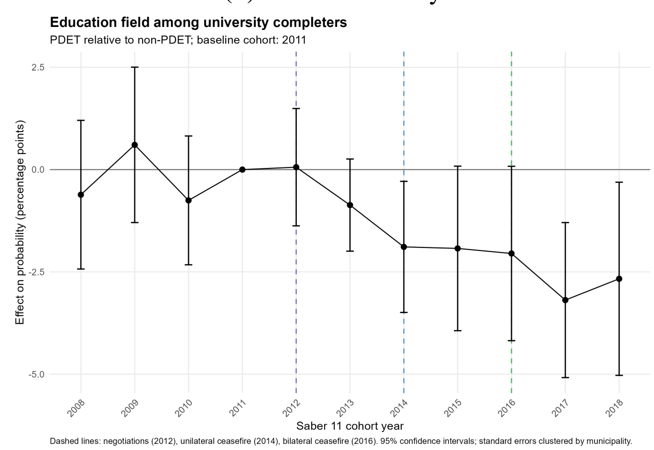
(a) Studied outside home municipality



(b) Public university



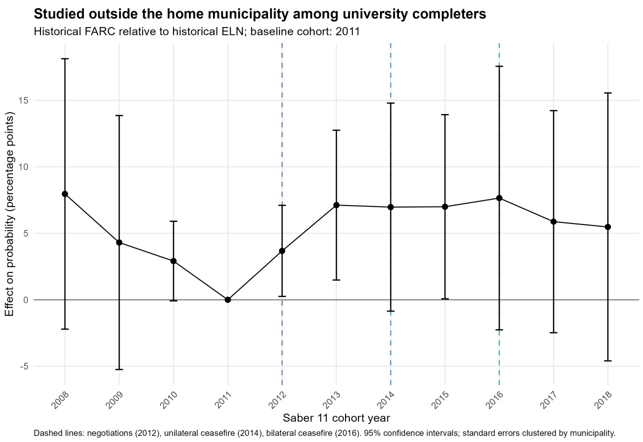
(c) STEM field



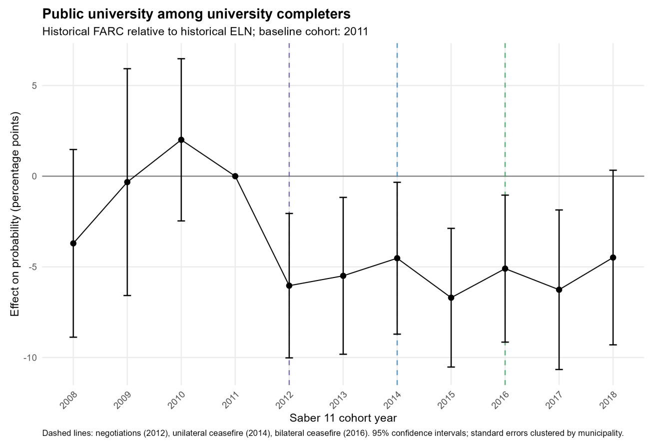
(d) Education field

Figure 11: Composition among PDET-origin university completers

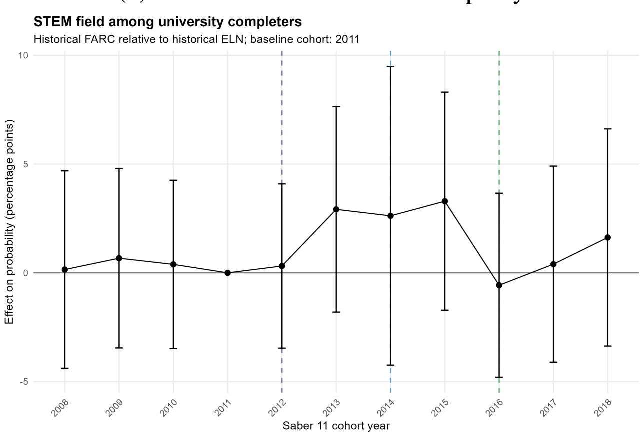
Notes: Effects are percentage points relative to the 2011 cohort. Pre-trend p values are 0.409 for migration, 0.623 for public university, 0.177 for STEM, and 0.399 for education.



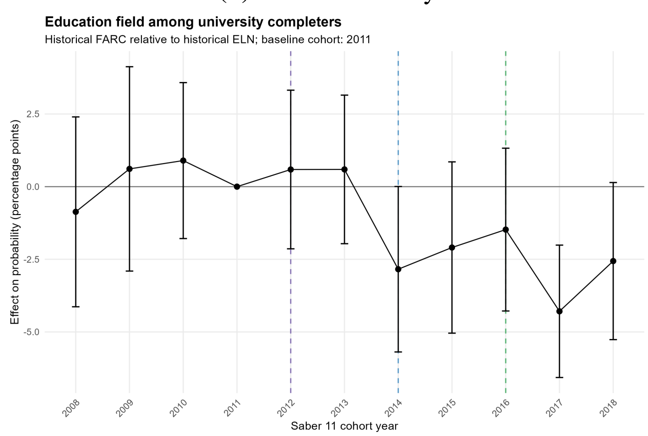
(a) Studied outside home municipality



(b) Public university



(c) STEM field



(d) Education field

Figure 12: Composition among historical FARC and ELN university completers

Notes: Effects compare historical FARC with historical ELN municipalities and are percentage points relative to 2011. Pre-trend p values are 0.161 for migration, 0.157 for public university, 0.987 for STEM, and 0.701 for education.

C Conflict Outcomes and Robustness

Figure 13 compares the pooled negotiation and post-agreement estimates for the conflict index, core event rate, homicide, and extortion. The FARC versus ELN decline is present under both policy windows and is robust to replacing the standardized index with the log population-normalized event rate. PDET conflict reductions are concentrated after the agreement, while extortion and homicide illustrate that broader public safety need not follow guerrilla-related violence one for one.

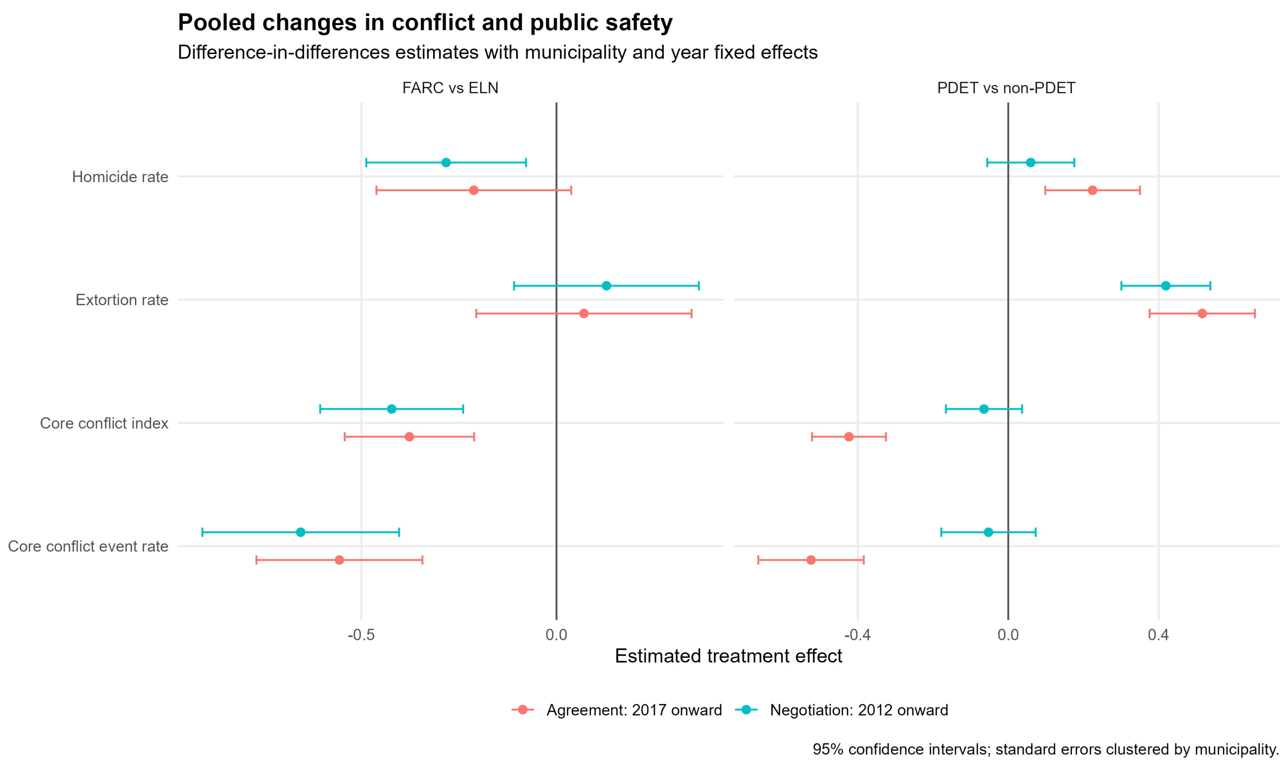


Figure 13: Pooled conflict and public-safety regressions

Notes: Points report treatment-by-policy-period coefficients from municipality-year models with municipality and year fixed effects. Equal municipality weights are used. Horizontal bars are 95 percent confidence intervals from municipality-clustered standard errors. Negotiation models define 2012 to 2018 as post; agreement models define 2017 to 2018 as post.

Table 3: Main conflict regressions

Design	Outcome	Policy period	Estimate (SE)
PDET vs non-PDET	Core conflict index	Negotiation, 2012 onward	-0.064 (0.052)
PDET vs non-PDET	Core conflict event rate	Negotiation, 2012 onward	-0.053 (0.064)
PDET vs non-PDET	Core conflict index	Agreement, 2017 onward	-0.423 (0.050)
PDET vs non-PDET	Core conflict event rate	Agreement, 2017 onward	-0.524 (0.072)
FARC vs ELN	Core conflict index	Negotiation, 2012 onward	-0.422 (0.094)
FARC vs ELN	Core conflict event rate	Negotiation, 2012 onward	-0.656 (0.129)
FARC vs ELN	Core conflict index	Agreement, 2017 onward	-0.377 (0.085)
FARC vs ELN	Core conflict event rate	Agreement, 2017 onward	-0.556 (0.109)

Notes: All models include municipality and year fixed effects. Standard errors in parentheses are clustered by municipality. Dependent variables are defined in Code 10.

Notes: Each entry reports a treatment-by-post coefficient followed by its municipality-clustered standard error in parentheses. All specifications contain municipality and year fixed effects. The core index averages standardized log population-normalized rates for the five event categories with complete 2007 to 2018 coverage.

C.1 Conflict-variable construction

The core index includes subversive actions, attacks on bridges and roads, kidnapping, terrorism, and oil-pipeline bombing. Attacks on the security forces are excluded from the index because that file begins in 2010. Homicide, extortion, and land invasion are treated as broader public-safety measures. Cocaine observations are excluded from the main conflict regressions because the source records enforcement quantities and does not measure hectares cultivated, production, or territorial exposure to coca markets. For each source, a zero is assigned only to municipality-years between the first and last observed source years. Population denominators come from the CEDE municipal characteristics panel.

D Uncertainty Outcomes and Measurement Limits

Figure 14 separates the two continuous timing measures and reports effects in months. Total time to Saber Pro rises in PDET territories, while excess time relative to semester does not. This distinction is why the main text does not equate longer progression with enrollment interruption.

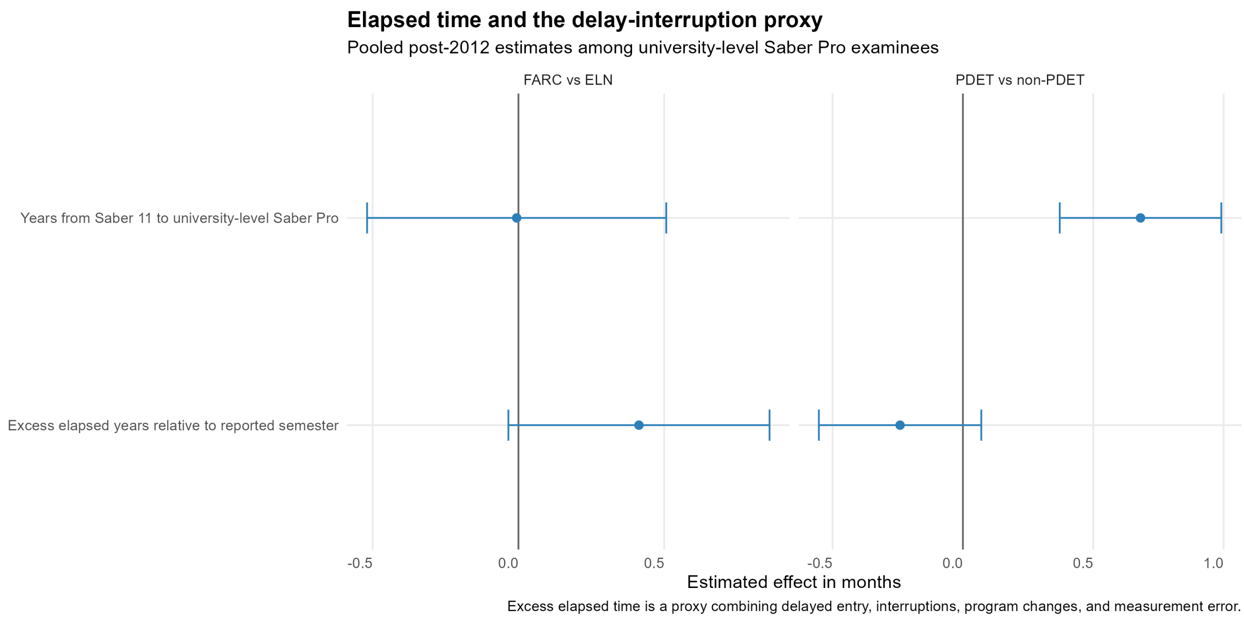


Figure 14: Elapsed time and the delay or interruption proxy

Notes: Pooled post-2012 models condition on reaching university-level Saber Pro within six years. Excess elapsed time subtracts one-half of the semester reported at Saber Pro from years since Saber 11. It is a proxy combining delayed entry, interruptions, program changes, and measurement error. Models include municipality, Saber 11 cohort, and where appropriate Saber Pro period fixed effects.

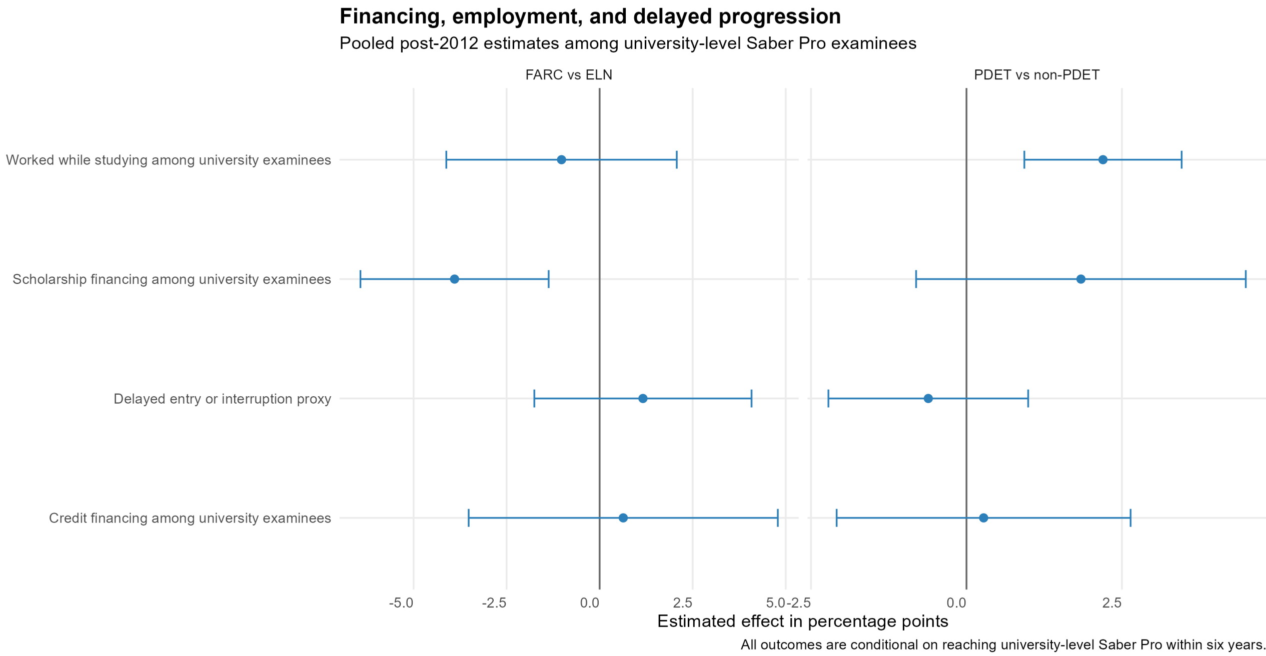


Figure 15: Financing, employment, and delayed progression among university examinees

Notes: Outcomes are measured among students reaching university-level Saber Pro within six years and are expressed in percentage points. Financing variables describe whether tuition was paid with credit or scholarship resources. Employment records whether the examinee worked during the previous week. The delay or interruption indicator equals one when excess elapsed time is at least one year.

Table 4: Pooled uncertainty regressions

Design	Outcome	Estimate	SE	Pre-trend p
FARC vs ELN	Credit financing among university examinees	0.634	2.119	0.695
FARC vs ELN	Delayed entry or interruption proxy	1.163	1.489	0.400
FARC vs ELN	Excess elapsed years relative to reported semester	0.034	0.019	0.631
FARC vs ELN	Saber Pro within four years	0.235	0.436	0.662
FARC vs ELN	Saber Pro within five years	0.358	0.624	0.469
FARC vs ELN	Saber Pro within six years	0.111	0.807	0.332
PDET vs non-PDET	Credit financing among university examinees	0.276	1.206	0.952
PDET vs non-PDET	Delayed entry or interruption proxy	-0.614	0.820	0.336
PDET vs non-PDET	Excess elapsed years relative to reported semester	-0.020	0.013	0.296
PDET vs non-PDET	Saber Pro within four years	-0.124	0.283	0.000
PDET vs non-PDET	Saber Pro within five years	0.725	0.338	0.010
PDET vs non-PDET	Saber Pro within six years	1.329	0.395	0.274

Notes: Binary outcomes are reported in percentage points; excess elapsed time is reported in years. All models include municipality and Saber 11 cohort fixed effects and control for cohort-standardized Saber 11 mathematics and reading scores. Conditional Saber Pro outcomes include examination-period fixed effects. Pre-trend tests jointly evaluate the 2008 to 2010 event coefficients.

Table 5: Observed and unobserved uncertainty margins

Margin	Status in ICFES data	Interpretation
Saber Pro within 4, 5, or 6 years	Directly observed from linked exam dates	Speed to near-completion, not speed to first enrollment
Years from Saber 11 to Saber Pro	Directly observed	Total elapsed progression time among examinees
Current program semester	Self-reported in most Saber Pro years	Program position at the exam; not a complete enrollment history
Excess elapsed time	Constructed proxy	Delayed entry, interruption, transfer, program change, or reporting differences
Credit and scholarship use	Directly reported at Saber Pro	Financing composition among selected near-completers
Work while studying	Directly reported at Saber Pro	Employment in the week before the exam
Applications	Not observed	Requires application records from institutions or a centralized system
Initial enrollment and interruptions	Not observed	Requires SNIES or SPADIES student histories linked to Saber 11

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